CrunchBite — Kitchen & Operations Team: Technology Context

Services Owned by This Team

Service	Responsibility	Technology
Kitchen Service	KDS ticket delivery, station routing, bump/recall, SOS timer calculation	Lambda + WebSocket API Gateway + DynamoDB
Store Config Service	Store hours, delivery radius, price overrides, feature flags per store	Lambda + DynamoDB
Analytics Service	Real-time SOS metrics, item-level sales, franchise roll-up, Snowflake feed	Kafka → Snowflake Kafka Connector + Lambda
Catering Service	Group order management, coordinator assignment, lead-time validation	Lambda + Aurora PostgreSQL 15

Frontends Owned by This Team

Client	Technology	Notes
KDS App (station display)	Electron 30 + React 18	Windows 10 IoT hardware; WebSocket connection to Kitchen Service; offline-tolerant with local queue
KDS Bump Screen (handheld)	Kotlin (Android)	Tablet-based handheld bump screens for kitchen staff; MQTT for sub-second real-time updates
Manager Dashboard	React 18 SPA	Tablet/desktop; 86-ing controls, SOS dashboard, void/refund, end-of-day report; Apollo Client + GraphQL BFF
Franchise Dashboard	React 18 SPA + Apollo Client	Multi-store roll-up; data sourced from Snowflake via GraphQL BFF; Tableau embeds for trend charts

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Real-Time Architecture (Kitchen)

WebSocket + MQTT

- **Kitchen Service** maintains WebSocket connections to the KDS desktop app (Electron) via AWS WebSocket API Gateway
- **MQTT broker** (AWS IoT Core) handles real-time communication to Kotlin Android handheld bump screens
  - MQTT topic structure: `store/{storeId}/station/{stationName}/ticket`
  - QoS level 1 (at-least-once delivery) for ticket events; idempotent by `ticketId`
- Station routing: when an order is placed, Kitchen Service determines which stations are involved based on item-to-station mapping and publishes to the relevant MQTT topics and WebSocket channels simultaneously

SOS Timer

- SOS start time = timestamp of `order.placed` Kafka event consumed by Kitchen Service
- SOS end time = timestamp when all items on the expediter bump screen are bumped for that order
- Real-time SOS per order is stored in DynamoDB (TTL 24 hours)
- Aggregated SOS metrics (store average, rolling 15-min average) published to Snowflake via Kafka Connector every 60 seconds

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Key Technical Constraints

Kitchen Service

- Must deliver a KDS ticket to the relevant station screen(s) within **1 second** of receiving the `order.placed` event from Kafka
- Offline resilience: if WebSocket connection drops, the KDS Electron app stores incoming tickets in a local queue and replays them when the connection is restored; no tickets should be lost
- Bump operations are idempotent: bumping an already-bumped item is a no-op (prevents duplicate bumps from network retries)

Handheld Bump Screens (Kotlin + MQTT)

- Android 10+ tablets with AWS IoT Core device certificates for MQTT authentication
- Device certificates rotated quarterly via AWS IoT Device Management
- Kotlin app uses MQTT QoS 1; reconnect strategy: exponential backoff up to 30 seconds

Analytics Service

- Kafka consumer group reads from `order.placed`, `order.ready`, `item.86d`, and `shift.opened` topics
- Data is transformed and loaded into Snowflake via the Kafka Snowflake Connector
- Snowflake tables power Tableau dashboards; refresh cadence is near-real-time for SOS and 15-minute batches for sales aggregates
- Franchise dashboards accessed via a GraphQL BFF that queries Snowflake (read-only, row-level security by store ID)

External Integrations (this team)

Integration	Protocol	Direction	Notes
Oracle MICROS POS	MICROS REST API + TCP socket	Kitchen Service → MICROS	Order sync, receipt printing, drawer management
Epson KDS hardware	WebSocket (KDS app)	Kitchen Service → Epson display	Station ticket delivery
AWS IoT Core (MQTT)	MQTT	Kitchen Service → Android tablets	Handheld bump screen updates
Infor CloudSuite F&B	REST API (batch nightly)	Menu Service ← Infor	Inventory and recipe updates (shared with Digital Ordering)

Medallia	REST API (post-transaction trigger)	Analytics Service → Medallia	NPS survey trigger after every completed order
Snowflake	Kafka Connector	Analytics Service → Snowflake	SOS metrics, sales data, franchise roll-up
Tableau	Snowflake JDBC	Tableau ← Snowflake	Franchise and FBC dashboards

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Non-Functional Requirements (this team)

Category	Requirement
KDS ticket latency	Ticket must appear on correct station screen(s) within 1 second of order.placed event
SOS accuracy	SOS timer accuracy ±5 seconds; timestamps from Kafka event (not client-side)
MQTT delivery	QoS 1 (at-least-once) on all kitchen events; duplicate handling via idempotent ticket ID
Dashboard refresh	SOS real-time (60-second aggregation); sales dashboard 15-minute batches; franchise roll-up 15 minutes
Offline resilience	KDS Electron app must queue tickets locally during WebSocket disconnection; no ticket loss
Availability	Kitchen Service 99.99% during store trading hours; a KDS outage directly stops food production